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Long-term dynamics and population viability in one of the last populations of the endangered *Spiranthes spiralis* (Orchidaceae) in the Netherlands

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ABSTRACT

During the last decades, most orchid species in much of Western Europe have suffered significant declines and the long-term survival of the remaining populations remains to a large extent uncertain. In particular, populations at range margins may be more prone to extinction than more central populations, as the former tend to be small and isolated, occur in ecologically marginal habitats and have a lower per-capita reproductive rate. In this study, we investigated the long-term dynamics and population viability of a population at the margin of its range of *Spiranthes spiralis* in the Netherlands. At present, only 2 out of 40 previously known populations persist. Individual plants were monitored for 24 years and their life span, flowering frequency and vegetative growth were determined. Individual plants showed large temporal variation in sexual and vegetative growth among years. The proportion of flowering plants varied from 0 (no plants were flowering) to 100 (all plants were flowering). Vegetative growth, on the other hand, increased when the number of individuals decreased. Dormancy was present, but occurred only in a few individuals. Using a non-structured population viability model, future prospects of this species were assessed. Calculation of extinction probabilities and estimated times to extinction using the diffusion approximation model showed that the species had a relatively high probability (79%) of surviving the next 20 years, whereas the median time to extinction was forty years. However, because 95% confidence intervals of the population growth included 1, we suggest that continued monitoring and additional genetic research are needed to assess the long-term viability of this species.

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1. Introduction

During the last decades, most orchid species in much of Western Europe have suffered significant declines, most likely as a result of land use changes, habitat deterioration and fragmentation (e.g. Jacquemyn et al., 2005a; Kull and Hutchings, 2006), whereas the viability of the remaining populations remains to a large extent uncertain. In particular, populations at range margins may be more prone to extinction than central populations because they tend to be small and isolated, often occur in ecologically marginal habitats and have a low per-capita reproductive rate (Lawton, 1993; Hoffman and Blows, 1994; Lesica and Allendorf, 1995; Sagarin and Gaines, 2002). At range margins, adequate

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management is therefore most of the times a prerequisite to maintain viable plant populations (see for example Oostermeijer et al., 2003).

The long-term survival of plant populations is determined by their vital rates, i.e. reproduction, recruitment, growth, and survival. However, the relative importance of these rates may differ in different environments. Under unfavorable conditions, individual plants may invest relatively more resources into clonal growth as a means to survive unfavorable habitat conditions (Jacquemyn et al., 2005b, 2006). Under such harsh conditions, sexual recruitment may be suppressed, locally less adapted genotypes may be outcompeted by better-adapted genotypes (Hartnett and Bazzaz, 1985), and only one or a few genotypes may survive (Schaal and Leverich, 1996; Klein and Steinger, 2002).

Sexual reproduction and seedling recruitment, on the other hand, are accompanied by genetic recombination and hence by the addition of new genotypes, which may allow adaptation to changing environmental conditions (Wise et al., 2002). Additionally, seed dispersal allows a species to travel greater distances and thus to escape unfavorable habitat conditions and to occupy a wider range of suitable microsites than dispersal by vegetatively produced propagules (Loehle, 1987; Gardner and Mangel, 1999; van Kleunen et al., 2001).

Most terrestrial orchid species possess the capability to reproduce sexually as well as clonally (Klimeš et al., 1997), although little is known about the relative importance of clonal growth and sexual reproduction in determining demography and population viability of orchid species. The few available studies indicate that sexual recruitment is prevalent among orchid species (but see Brzosko et al., 2002; Machon et al., 2003; Chung et al., 2004). This is surprising because most orchids are pollinator limited, recruitment through sexual reproduction is very unpredictable due to symbiosis with fungi and because sexual reproduction may come at a fitness cost (e.g. Snow and Whigham, 1989; Primack and Hall, 1990). In case trade-offs between fruit set and flowering frequency constrain the lifetime fitness of orchid populations (Calvo, 1993), clonal growth may be an efficient means to survive unfavorable habitat conditions.

In this study, we investigated the long-term dynamics and population viability in a population at the margin of its range of the extremely rare, long-lived, perennial orchid *Spiranthes spiralis* in the Netherlands. Despite the fact that appropriate management has prevented extinction during the last 25 years, the long-term survival largely remains uncertain. The aims of this study were, firstly, to provide insights in the demography of this species, and secondly, to assess its future prospects. The meet these aims, demographic data were collected for 24 years in a large permanent plot. More specifically, we investigated how clonal growth and sexual reproduction varied among years, and how clonal growth affected survival. Finally, a population viability analysis using the diffusion approximation (Dennis et al., 1991) was performed to assess future prospects of the species and to estimate extinction probabilities of the studied population within specific time horizons.

2. Material and methods

2.1. Species

Spiranthes spiralis (L.) Chevall. is a small, long-lived perennial orchid that is widely distributed in Southern Europe, in the Mediterranean region and in coastal areas of North Africa (Ziegenspeck, 1936; Tutin et al., 1980). The species reaches the limits of its northwestern distribution in the southern part of the United Kingdom and The Netherlands. The species has suffered significant declines in much of Western Europe, most likely being the result of habitat loss and deterioration (Jacquemyn et al., 2005a; Kull and Hutchings, 2006). In the Netherlands, *S. spiralis* is listed as endangered: out of around 40 populations that were present before 1950, only two still exist today (Willems and Lahtinen, 1997; see also Kreutz and Dekker (2000) for past and present distribution maps).

S. spiralis typically occurs in relatively dry, nutrient-poor meadows or calcareous grassland (Weeda et al., 1994; Kreutz and Dekker, 2000). It typically occurs in open grazed situations with a constant land-use. Individual plants survive during many years and may consist of several rosettes and one or several small belowground tubers per plant (Wells, 1967; Willems and Lahtinen, 1997; Machon et al., 2003). New tubers are formed during the winter and spring, and leaves as well as inflorescences emerge in late summer. Leaves are small (between 1 and 5 cm long and 0.5 and 1.5 cm wide) and die off in late spring or early summer. *S. spiralis* flowers in August and September. The height of the inflorescence varies between 5 and 25 cm and it may bear up to 25, small, white flowers that are arranged in a spiral on the upper half of the flowering stalk. Flowers are self-compatible, produce nectar and are mainly pollinated by bumblebees and bees (van der Cingel, 1995; Willems and Lahtinen, 1997). Spontaneous self-pollination does not occur. In October to November the capsules become ripe. Fruit set varies between 0% and 78% (mean: 35%) (Willems and Lahtinen, 1997). The seeds are wind-dispersed, but most seeds fall in the direct neighborhood of the mother plant (Machon et al., 2003). Vegetative multiplication occurs by the growth of a lateral bud on the underground stem, the new plant forming its own tubers, and eventually the connection between the mother plant and vegetatively produced plants withers away (Wells, 1967). In this way, small groups of plants are formed, which are easily distinguished from plants that have arisen from seed.

2.2. Study site

This study was conducted from 1981 to 2004 in the Berghofweide Nature Reserve in the Netherlands (50°50′N; 5°53′E). This site is a species-rich calcareous grassland located on a south-facing slope (inclination 15°, altitude 170 m). The subsoil consists of a mixture of gravel deposits and weathered chalk products. The pH of the loamy topsoil is high (6.0–7.5). Before 1980, both manure and artificial fertilizer were applied yearly to the site, but in limited quantities. From 1980 onwards, this practice was abandoned and management nowadays consists of yearly mowing and hay removal in summer (July) and grazing in late autumn or early winter by sheep.

2.3. Data collection

From 1981 till 2004, a total of 201 genets were monitored annually in a permanently marked plot ($8 \times 8\ m^2$) (see Willems, 1989 for more details). Every year, the location of each genet was located and mapped. In this way, it was possible to return to the same plant in successive years, which led to accurate, annual counts of individuals in the population. In some cases, plants could not be retrieved in year t, although they were observed in year $t-1$ and $t+1$. This suggests that dormancy in this species may occur. However, in most cases it was more likely that due to grazing plants had already disappeared at the period of sampling. Further, the number of leaves, the number of inflorescences and the number of rosettes was also determined annually for each genet.

2.4. Data analysis

Annual changes in the number of flowering and non-flowering individuals, the number of individuals that were not detected in year t despite their presence in year $t-1$ and $t+1$, and the total number of genets were assessed by plotting these numbers against year. In order to visualize the relationship between year and clonal growth, the mean number of rosettes per genet in a given year was plotted against year. This allows examination of differences in clonal and sexual reproduction over years. Further, the lifespan of each individual (i.e. the number of years an individual was observed aboveground since it was observed for the first time) was determined and averaged for all individuals of a single cohort (i.e. all individuals that were observed for the first time in a given year). The average aboveground life span was plotted against year in order to determine variations in lifespan among cohorts. Finally, Spearman rank correlation was used to determine the relationship between flowering frequency and life span.

To assess future prospects of this species, we used a non-structured population viability analysis (PVA) based on counts (Dennis et al., 1991). To briefly review, the diffusion approximation model assumes that the natural logarithm of the population size is adequately approximated by a Wiener process with drift and originates from the observation that at the population level, stochastic age-structured models with no density dependence behave as a stochastic, discrete time model with exponential growth or decline:

$$N_{t+\tau} = N_t(\mu\tau + \varepsilon) \quad \text{where } \varepsilon \sim N(0, \sigma^2\tau). \tag{1}$$

The parameter μ determines the rate at which the median increases through time, whereas σ determines the rate of spread of the distribution, or the variability of population size at time $t+\tau$. When censuses are conducted yearly and there are no missing census points, the maximum likelihood estimates of μ and σ^2 are (Dennis et al., 1991):

$$\begin{aligned} \hat{\mu} &= \text{mean}[\ln(N_{t+1}/N_t)], \\ \hat{\sigma}^2 &= \text{var}[\ln(N_{t+1}/N_t)]. \end{aligned} \tag{2}$$

Based on these measures, the average value of the population growth rate and its confidence interval can be calculated as

$$\begin{aligned} &\hat{\lambda} = \exp\left[\hat{\mu} + (\hat{\sigma}^2/2)\right], \\ &\exp\left(\tilde{r} \pm z_{\alpha/2}\sqrt{\tilde{\sigma}^2\left[\frac{1}{t_q} + \frac{\tilde{\sigma}^2}{2(q-1)}\right]}\right), \end{aligned} \tag{3}$$

where $q+1$ is the number of annual censuses and $z_{\alpha/2}$ is the $100[1-(\alpha/2)]$th percentile of the standard normal distribution.

The probability that extinction has occurred at some time, can be approximated by the probability that $X(\tau) \geqslant x_d$ at some time $\tau \leqslant T$ where $x_d = \log(n_q/n_e)$ and n_e is the quasi-extinction threshold. This probability can be expressed as

$$\Pr[\tau \leqslant T] = \Phi(U - V) + \exp(2UV)\Phi(-(U+V)), \tag{4}$$

where Φ is the standard Normal cumulative distribution function, and

$$U = |\hat{\mu}|\sqrt{T}/\sigma \quad \text{and} \quad V = x_d/\sigma\sqrt{T}. \tag{5}$$

The most recent population count was used as initial population size (n_q), and two quasi-extinction thresholds (n_e) were used: 1 individual and 4 individuals. Dennis et al. (1991) have shown that the probability of extinction can be calculated as

$$\hat{\pi} = (n_e/n_q)^{2\hat{\mu}/\hat{\sigma}^2} \tag{6}$$

and the mean time to extinction as

$$\theta = x_d/|\hat{\mu}|. \tag{7}$$

The median time to extinction was derived from the cumulative distribution function. However, because estimates of extinction probability may be largely uncertain for large time horizons (Ludwig, 1999; Fieberg and Ellner, 2000; Meir and Fagan, 2000), we also calculated the extinction probability of *S. spiralis* within the next 20 years.

3. Results

The total number of plants within the permanent plot varied between 6 in 1981 and 79 in 1996 (Fig. 1a). From 1997 onwards, the number of plants decreased again to reach a low value of 18 in 2004. The percentage of flowering individuals fluctuated dramatically between years and varied between 0 (no individuals were flowering) and 100 (all individuals in flower) (Fig. 1b). Years with high flowering frequencies (1981, 1984, 1992, 2002) were mostly followed by years with low flowering frequencies. Only in few cases, plants were not observed at a given year despite their presence in the previous and the next year (Fig. 1a).

Measured over all plants, genet size varied between 1 and 6 rosettes, but most genets consisted of one or two rosettes (mean 1.4). Mean genet size per year varied considerably among years (range: 1.16–1.88): it was highest in the early years of monitoring, sharply decreased until 1999 and started to increase again (Fig. 2). Changes in the mean number of rosettes were associated with changes in the number of individuals in the population (Fig. 3). Low numbers of individuals were related to large genet sizes (indicated by high numbers of rosettes per genet). The individual lifespan of the different cohorts varied between 1.33 and 9 years, and was highest at the beginning of the study (Fig. 4). In addition, the individual life span was significantly and negatively related to flowering frequency ($r_s = -0.37$, $p < 0.001$, $n = 199$).

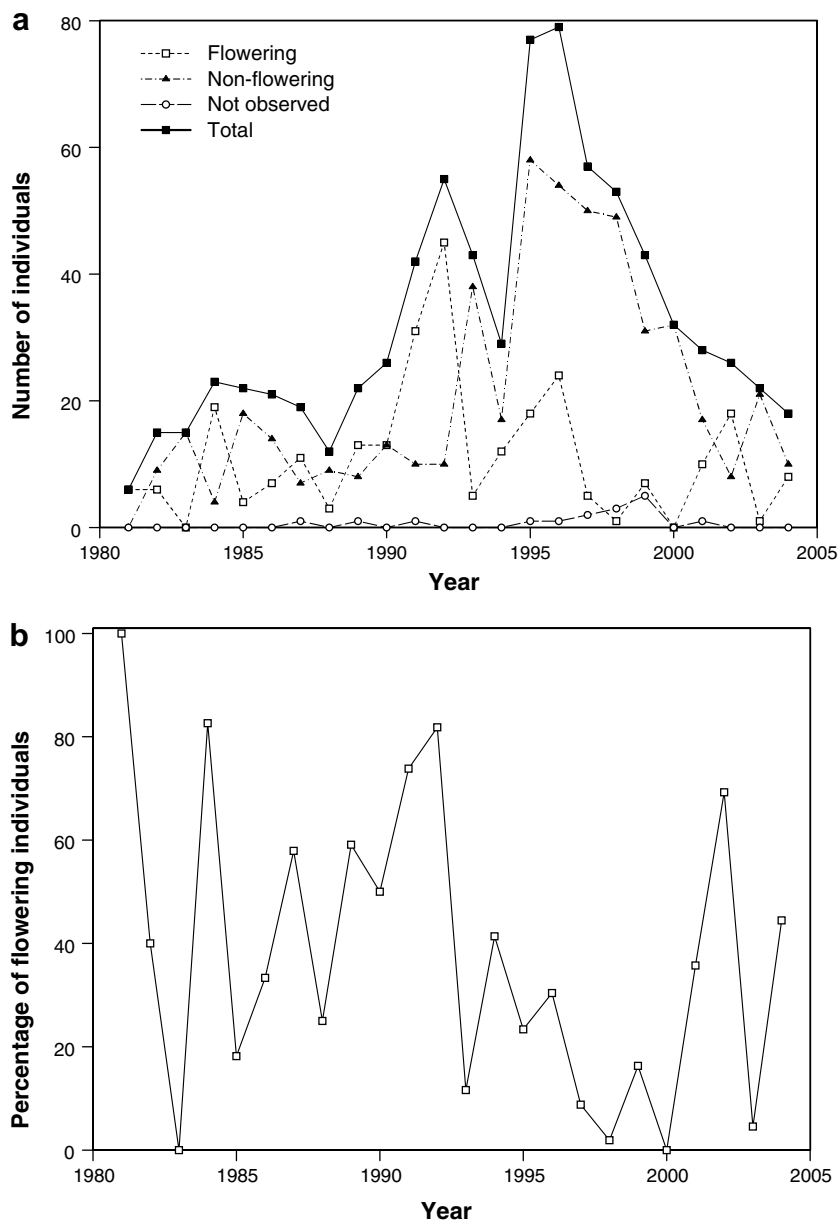

Fig. 1 – Population dynamics (a) and flowering frequencies (b) of *Spiranthes spiralis* between 1981 and 2004 at the Berghofweide Nature Reserve, the Netherlands.

Data from Fig. 1 yielded maximum likelihood estimates of μ and σ^2 of 0.014 and 0.154, respectively. The associated population growth rate λ was 1.133 (95% confidence interval: 0.959–1.339). Using a starting population size of 18 individuals (i.e. the population size of the last population census) and a pseudo-extinction threshold of 4 and 1 individuals, resulted in an ultimate extinction probability of 0.376 and 0.153, respectively. The mean time to extinction was 31 and 60 years, respectively. Fig. 5 shows the cumulative probability of extinction for an extinction threshold of 4 and 1 individual. From this figure, the median time to extinction can easily be derived for the different extinction thresholds. This was 15 and 40 years, respectively. However, it should be noted that μ was larger than 0. In this case, the extinction time CDF must be interpreted with caution, as it does not apply to all population realizations, but only to those that eventually reach the extinction threshold (Morris et al., 1999). However, the conditional extinction time CDF is still useful in that it allows estimating the total probability that a population will have gone extinct by a given time period. Using a short-term time horizon of 20 years, the total probability of reaching an extinction threshold of 4 and 1 individual is 0.21 and 0.13, respectively.

4. Discussion

4.1. *Flowering frequencies*

Our 24 years of data showed considerable variation in clonal reproduction, the proportion of plants flowering and life span. Large variations in flowering frequencies are typical among

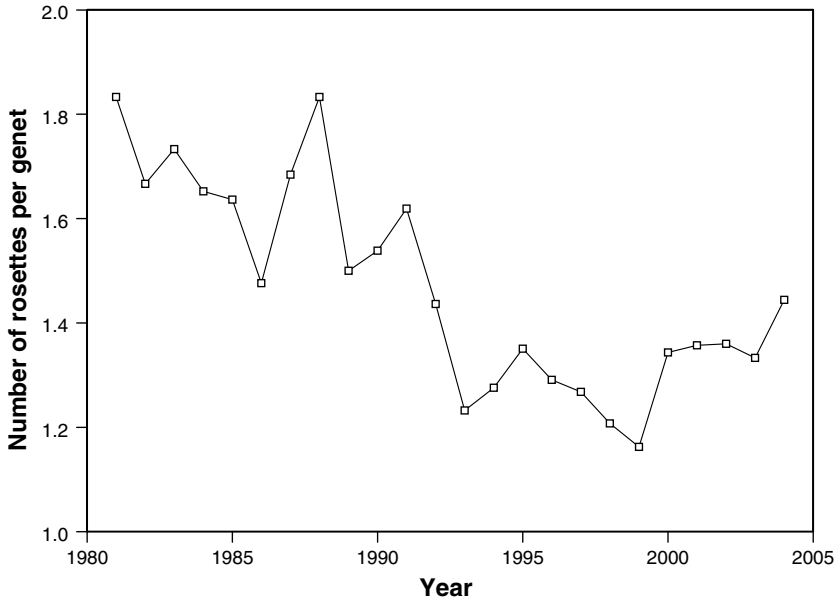

Fig. 2 – Genet size measured as the mean number of rosettes of *Spiranthes spiralis* individuals between 1981 and 2004.

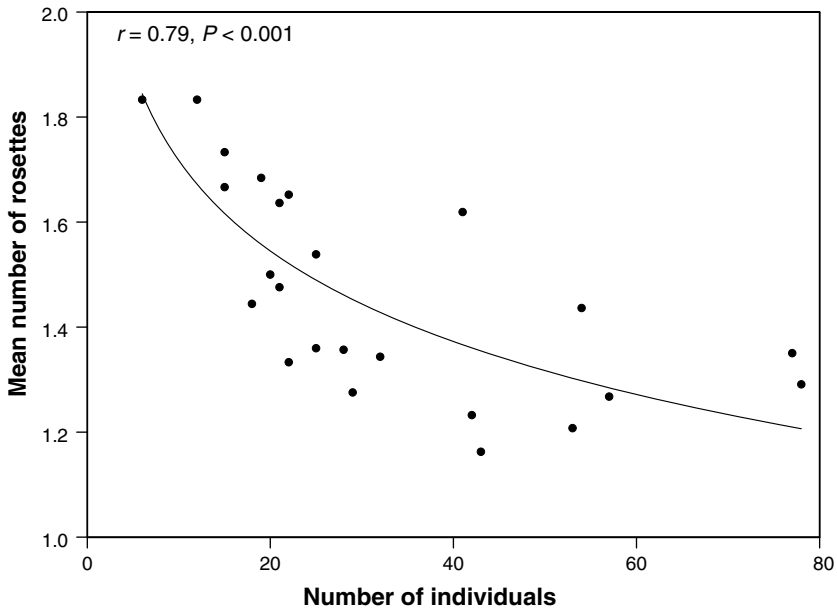

Fig. 3 – Relationship between the number of individuals and genet size (measured as the number of rosettes per genet) of *Spiranthes spiralis* determined for 24 years.

terrestrial orchid species (e.g. Hutchings, 1987; Shefferson et al., 2003). These may be related to costs of flowering, weather conditions or a combination of both. Most orchid species need to attain a certain threshold leaf area before they start flowering (e.g. Willems, 1982; Wells and Cox, 1989). In *S. spiralis*, flowering in a given year negatively affected the probability of flowering in the next year. Willems and Dorland (2000) indeed showed that on average more than 80% of all plants that flowered in year t were vegetative in year $t + 1$, suggesting that flowering is costly for this species. Rosettes also had a smaller leaf area in years of flowering comparing to year preceding or following flowering. However, a cost of flowering alone cannot explain the high synchrony in non-flowering individuals. Therefore, we can assume that weather conditions may have affected flowering frequencies of this species as well.

4.2. *Vegetative multiplication*

In orchids, clonal growth and the factors affecting clonal growth are still poorly understood. In this study, it was shown that genet size varied considerably among years (Fig. 2) and was significantly related to the number of individuals sampled. Because of the large number of small genets observed

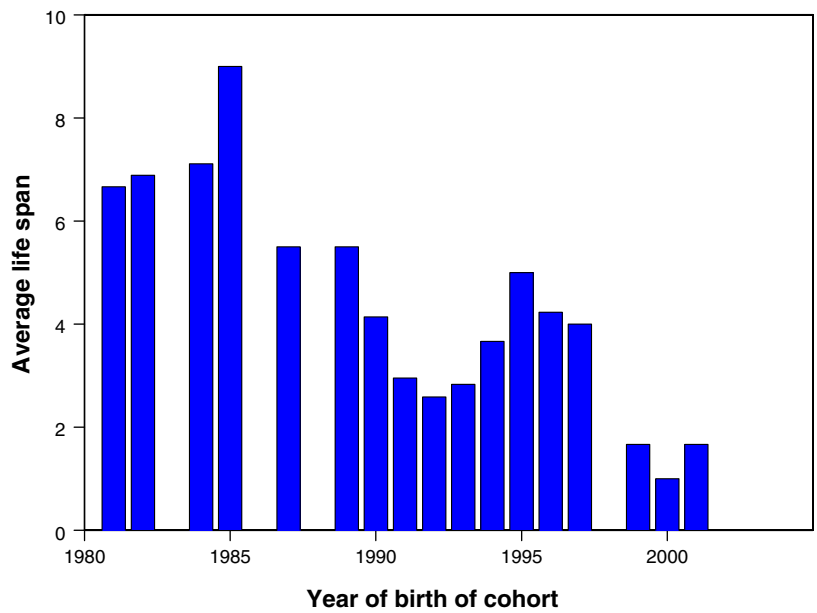

Fig. 4 – Average life span of *Spiranthes spiralis* estimated for different cohorts.

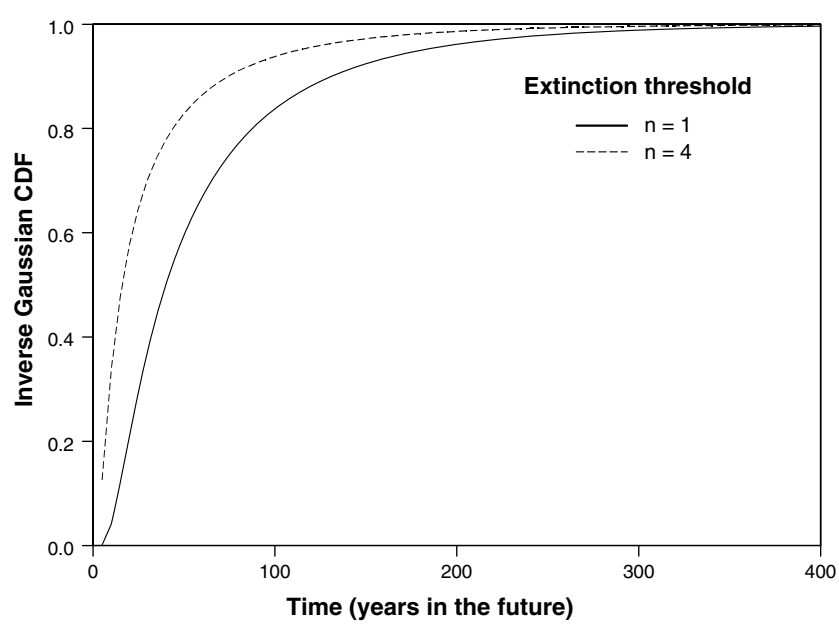

Fig. 5 – Cumulative distribution function for the probability of extinction for *Spiranthes spiralis*. The line shows the probability of extinction before a certain time in future.

between 1993 and 1999 and the parallel increase in the number of recruits, we may conclude that population expansion was mainly the result of sexual reproduction and associated high recruitment rates. On the other hand, in the first part of the study and at the end, when the number of genets started decreasing again, competition for light and space may have resulted in the survival of a few large individuals. Clonal growth has indeed been shown to be advantageous in unfavorable conditions (Jacquemyn et al., 2005, 2006). In addition, the average life span of *S. spiralis* (measured as the number of years an individual was observed aboveground since it was seen the first time) for different cohorts varied between 1.33 and 9 years, indicating substantial variation in longevity of individuals among cohorts. These figures are in accordance with those of Wells (1981), who found half-lives of cohorts of *S. spiralis* varying from 4.6 to 9.2 years.

4.3. *Population viability*

Despite the alluring simplicity of the diffusion approximation model, several assumptions (Morris et al., 1999) have to be met for the data to be reliable (e.g. accurate estimation of total population size, year-to-year variation reflects true environmentally driven variation, no large fluctuations caused by

extreme weather events or severe disturbances and absence of density-dependence). In our data set, most of these assumptions are met. First, measurement error was restricted to a minimum through a meticulous sampling scheme and repeated sampling within the same year. Moreover, the few dormant individuals that we observed were incorporated in the total counts, yielding more accurate figures of total sample size. Secondly, no extreme disturbances or catastrophes seem to have disturbed the populations so that changes in population size most likely reflect true environmental stochasticity. Thirdly, although density dependent reproduction may occur (Willems and Lahtinen, 1997), this probably does not largely affect the demography of the species. Moreover, recent simulation studies by Elderd et al. (2003) have shown that with long time series of census data, most important problems of the DA method are counterbalanced and that the method may be used as a valuable alternative to other population viability methods. Our analysis shows that the population has a fairly good chance of surviving the next 20 years, and that extinction is not very likely to occur within this period. However, inspection of the 95% confidence intervals around the population growth rate indicate growth rates smaller than 1, suggesting that further population decline is not unlikely.

4.4. *Future prospects*

At the moment, only two populations of *S. spiralis* remain in the Netherlands. Although from a demographic point of view the studied population has a good chance of surviving the next 20 years, these prospects may be too optimistic, as it was clearly shown that, despite stable management, the population has declined seriously during the last 10 years of monitoring. The exact causes for this decline are yet not known, but it may be that the viability of recruits may have been too low to survive the current conditions. Conservation management of the site, which was introduced in 1980 and consists of mowing in summer and winter grazing by sheep, aimed at creating a short sward during the period November to May and at avoiding overgrowing of the tiny rosettes by the surrounding vegetation. Before, the site was fertilized each year and no specific measures to maintain species diversity were undertaken. This change in management resulted in a massive growth of the population ten years later. However, given that the population has increased from a low number of individuals and given that relatively large clones were observed in the beginning of the study, inbreeding depression may have led to lower fitness of recruits (Fischer and Matthies, 1998; Keller and Waller, 2002). Indeed, pronounced clonal growth and small population sizes have been shown to substantially reduce the effective population size and increase the per-generation inbreeding in orchids (Chung et al., 2004). Relatively short life spans and high mortality rates may be indicative for this. In addition, small effective population sizes may also accelerate the rate at which potentially adaptive genetic variation is lost due to random genetic drift. Therefore, continued monitoring in combination with genetic analyses are needed to understand the exact causes of the decline of this species during the last years and to fully capture the observed changes in demographic parameters in order to propose management interventions that decrease the risk of extinction of one the last populations of *S. spiralis* in the Netherlands.

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